

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

Work, Power and Energy

LECTURE-3

PHYSICS

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Topics To Be Covered

- 1 Conservative Force ✓
- 2 Non-Conservative Force ✓
- 3 Vertical Circular Motion ✓
- 4 Collision
- 5 Elastic Collision — 0

Recap From 1st Lecture

Work

- mode of transfer of energy

$$W = \vec{F} \cdot \vec{x}$$

$$W = Fx \cos \theta$$

Work

W = +ive

W = -ive

W = 0

Work

Work done by constant



$$W = \vec{F} \cdot \vec{x}$$

Work done by variable force



$$W = \int F \cdot dx$$

Energy

- Capacity to do work

$$m \cdot E$$

K.E

+

P.E

$$K.E = \frac{1}{2}mv^2$$

(Derivation)

✓
✓
✓
↓
Detail
next
Lecture

Recap From 2nd Lecture

• Work-Energy Theorem

$$W = \Delta K \cdot \varepsilon$$

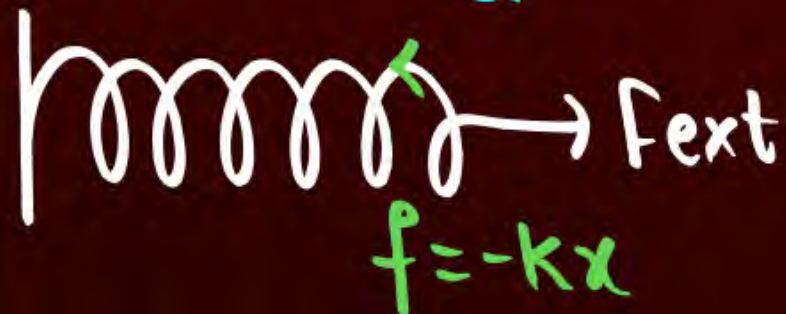
• P.E / U → Potential Energy

Zabardasti ✓

Relative position of particle

✓ Gravitation $P.E = mgh$

✓ Spring $P.E = \frac{1}{2} kx^2$



• WORK done by spring force

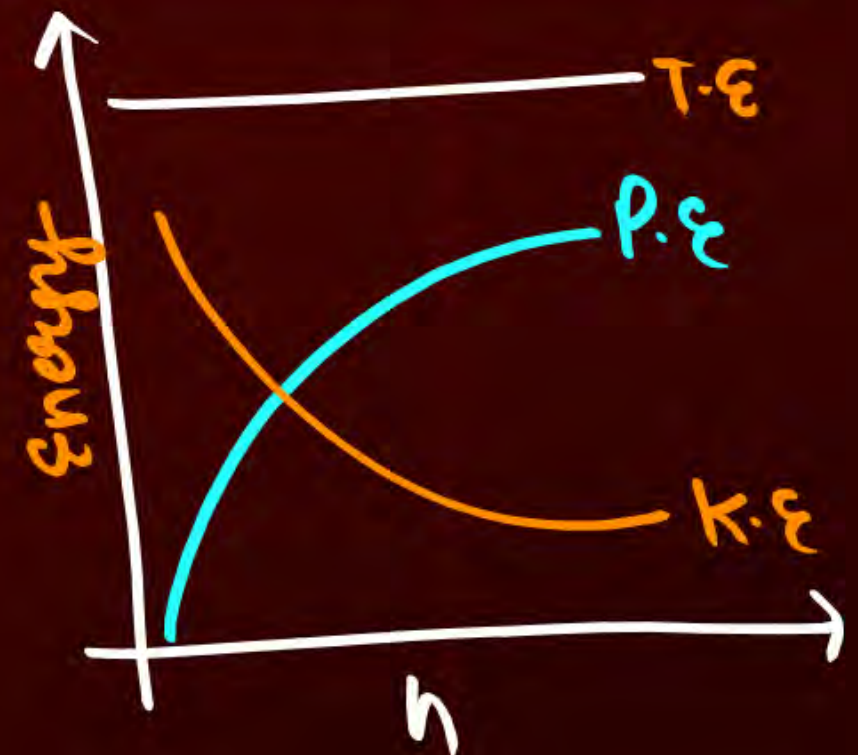
$$W = -\frac{1}{2} kx^2$$

Law of Conservation of Energy

$$T.E_i = T.E_f$$

$$m.E_i = m.E_f$$

$$100J \quad 100J$$



Conservative Force

⇒ Ahsaan hai

The force whose work done does not depend upon path

Example :- Gravitation Force ✓ mg
 ✓ Electrostatic force → 12th
 ✓ Spring force → 11th - 13

Path - ① work done → Displacement

$$W = \vec{F} \cdot \vec{x} = Fx \cos \theta$$

$$= mgh \cos 180^\circ$$

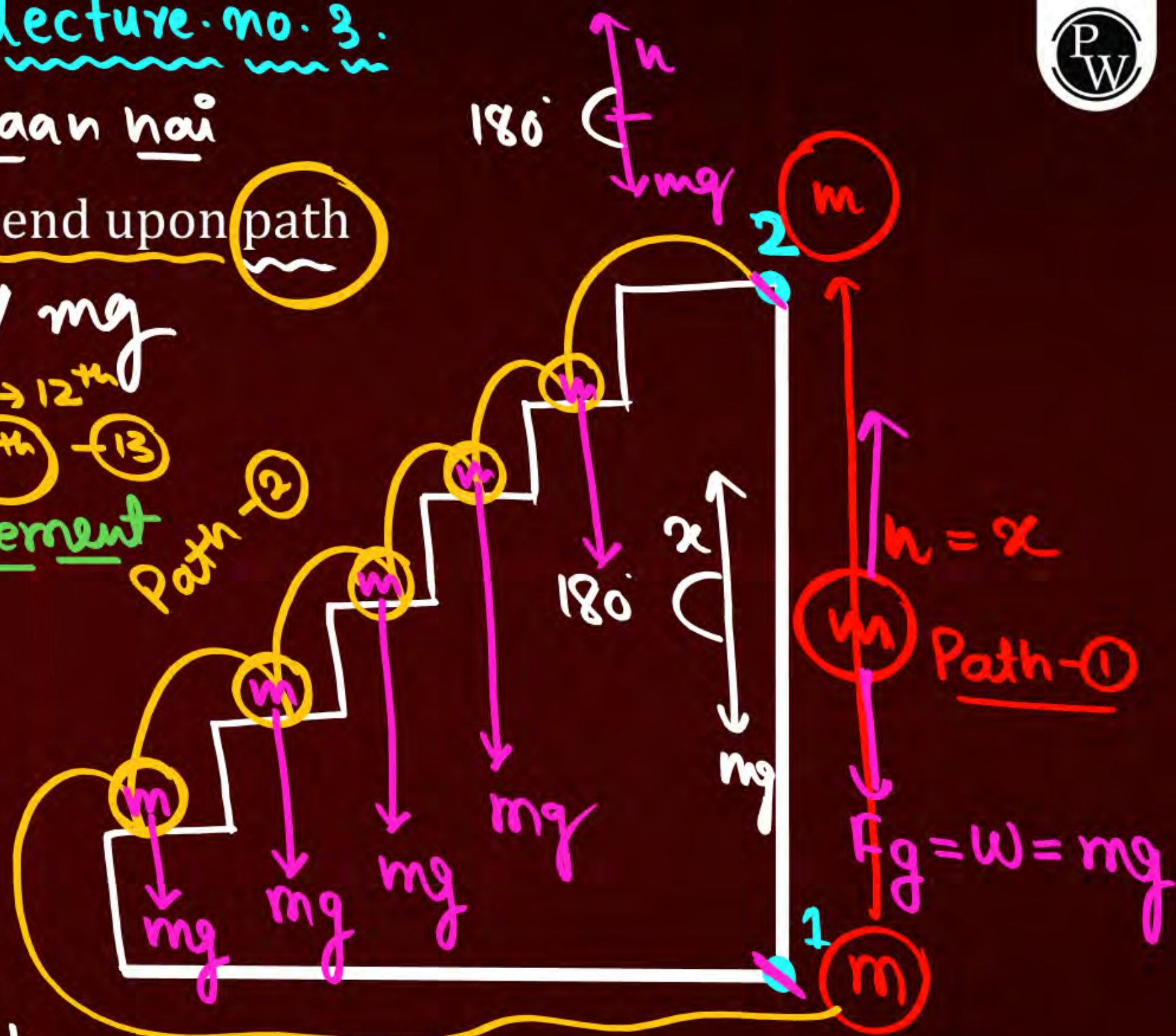
$$= mgh(-1)$$

$$W_1 = -mgh$$

Path - ②

$$W_2 = Fx \cos \theta$$

$$= mgh \cos 180^\circ = mgh(-1) = -mgh$$



12th

Electrostatic P.E $\vec{+q}$ $\vec{+q}$

✓ # Gravitation P.E $F = mg$ ✓

(F)

✓ # Spring P.E $f = -kx$ ✓

Neet

Note

Potential Energy is defined for conservative
force (o.i.i.)

NO $\rightarrow$ frictional P.E

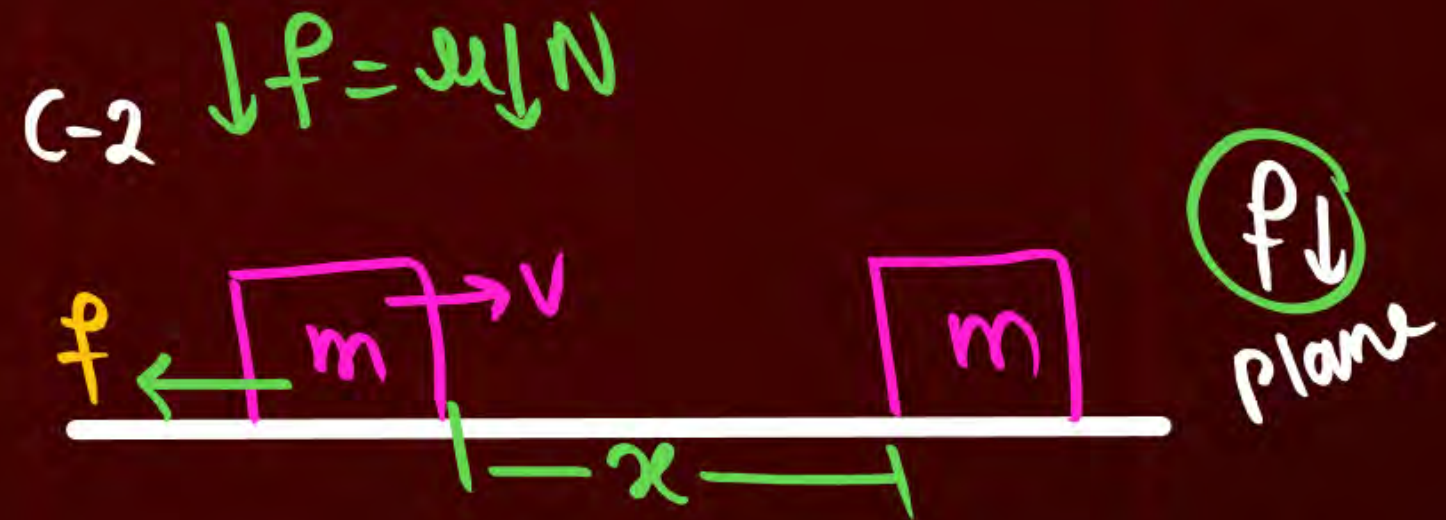
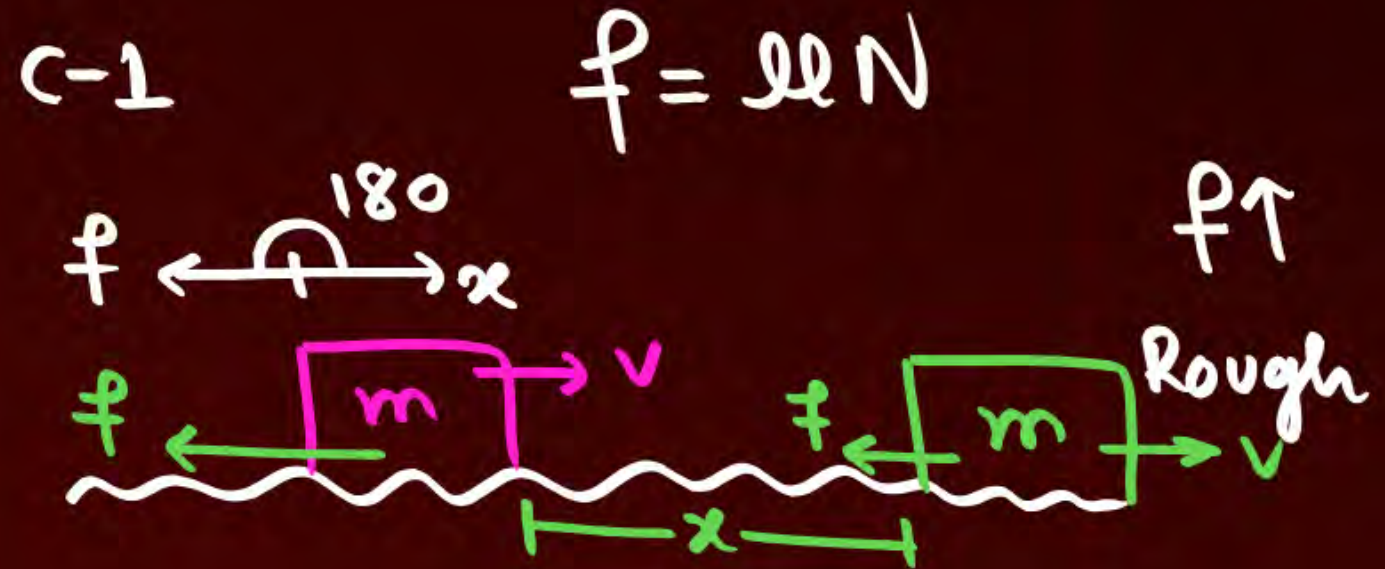
Non-Conservative Force

The force whose work depends on path

Example:- friction force

C-1 $W = Fx \cos \theta$
 $= fx \cos 180^\circ$
 $= fx(-1) = -fx$
 $\uparrow \textcircled{W_1} = -\mu N x$

C-2 $W_2 = fx \cos \theta$
 $W_2 = fx \cos 180^\circ$
 $W_2 = \mu N x (-1)$
 $\downarrow \textcircled{W_2} = -\mu N x$

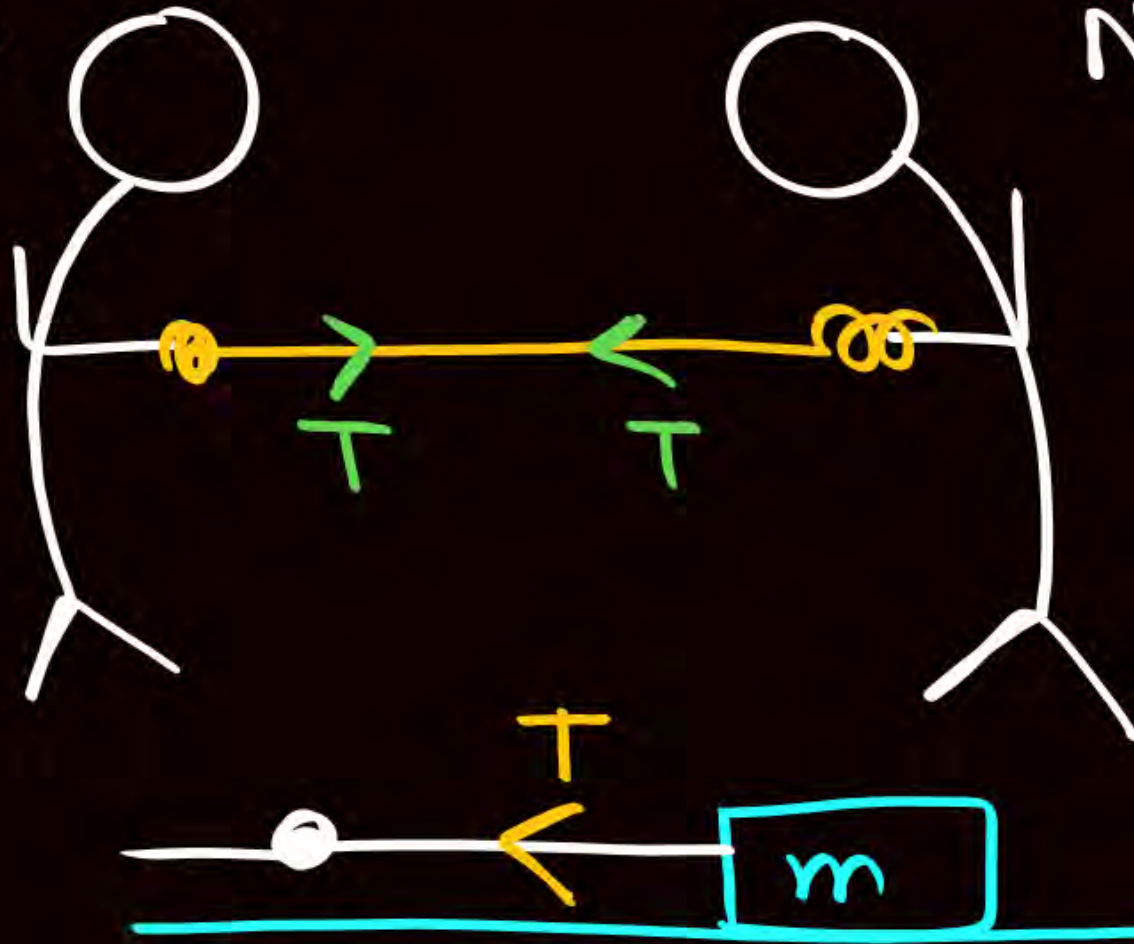


Tension :- (force)

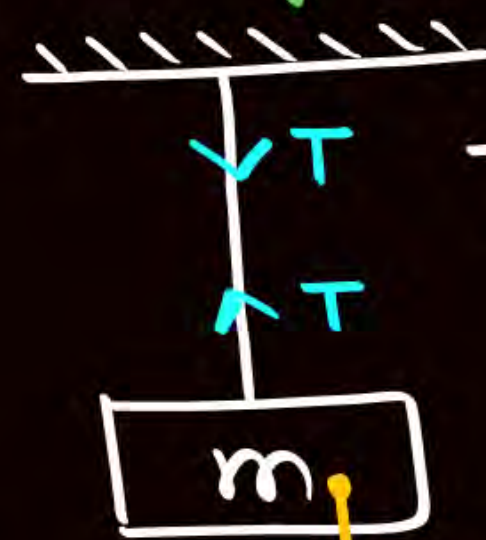
The force in a string (Rassi) when it is stretched.

Its direction is away from the Body

Bashir



Nazir



Dimension

$$T = f = \underline{MLT^{-2}}$$

2016
Board

Unit of Tension

Newton

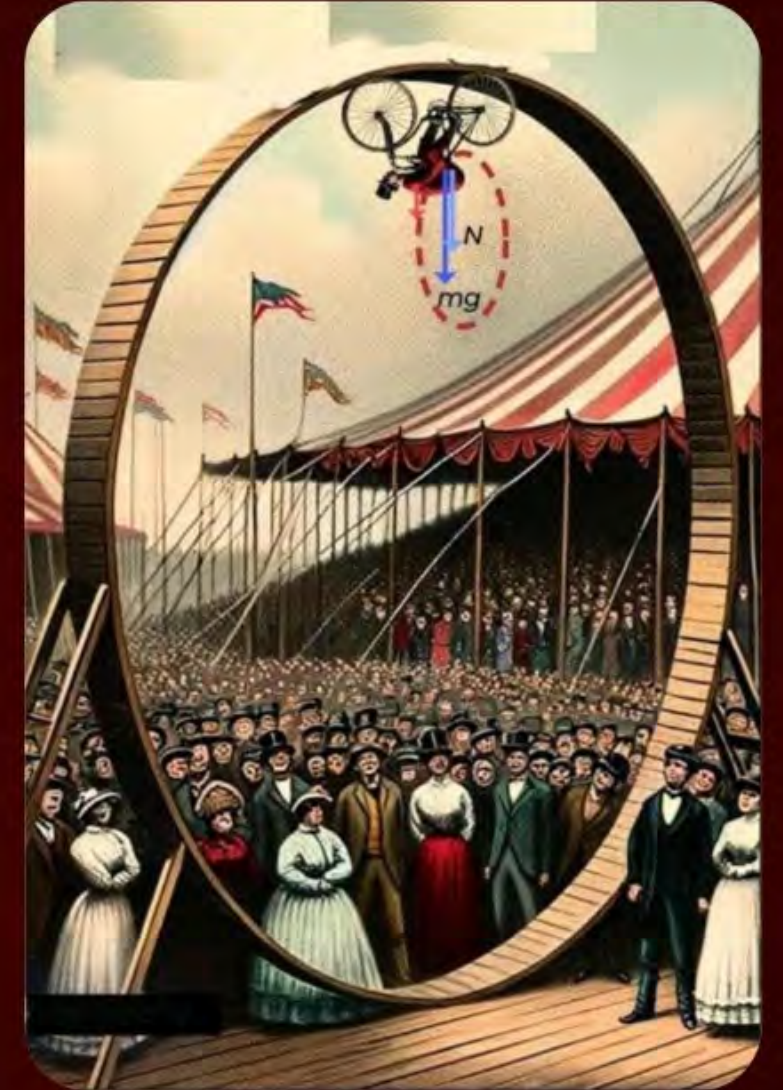
$w = mg$

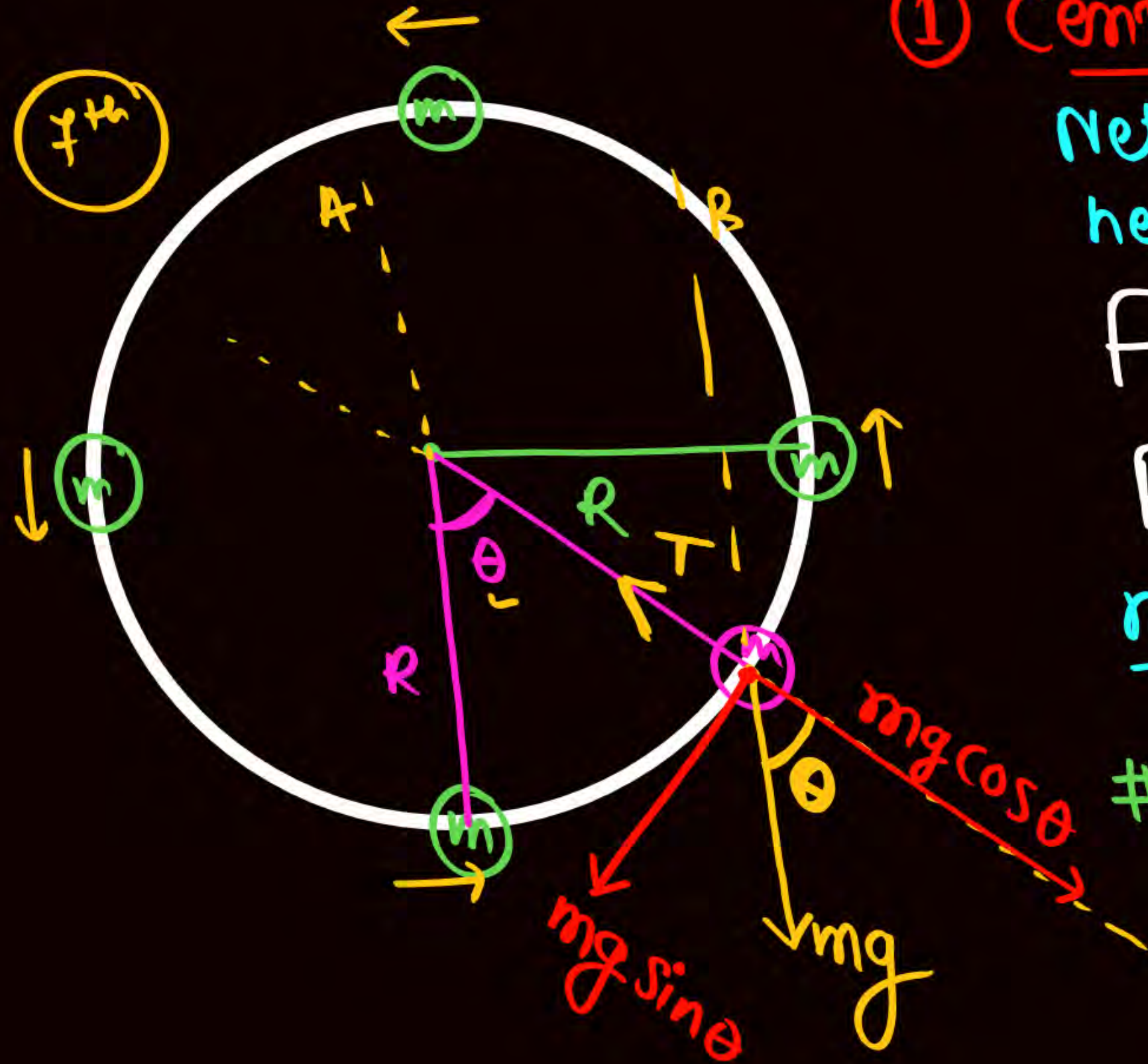
Vertical Circular Motion

When you move the object in a vertical Circle

Centripetal Force ✓ $F_c = \frac{mv^2}{r}$

Tension ✓





① Centripetal force

Net inward force which help in circular motion

$$F_{\text{inward}} = T - mg \cos \theta$$

$$F_c = T - mg \cos \theta \quad \text{--- ①}$$

$$\frac{mv^2}{R} = T - mg \cos \theta \quad \checkmark$$

Tension

$$\frac{mv^2}{R} + mg \cos \theta = T \quad \text{--- ②}$$

Collision

(Bohot Asson Karlega)

- Takrana ✓ collide
- When one object bombared over the other object
- During collision energy and momentum is transfer

$K.E$

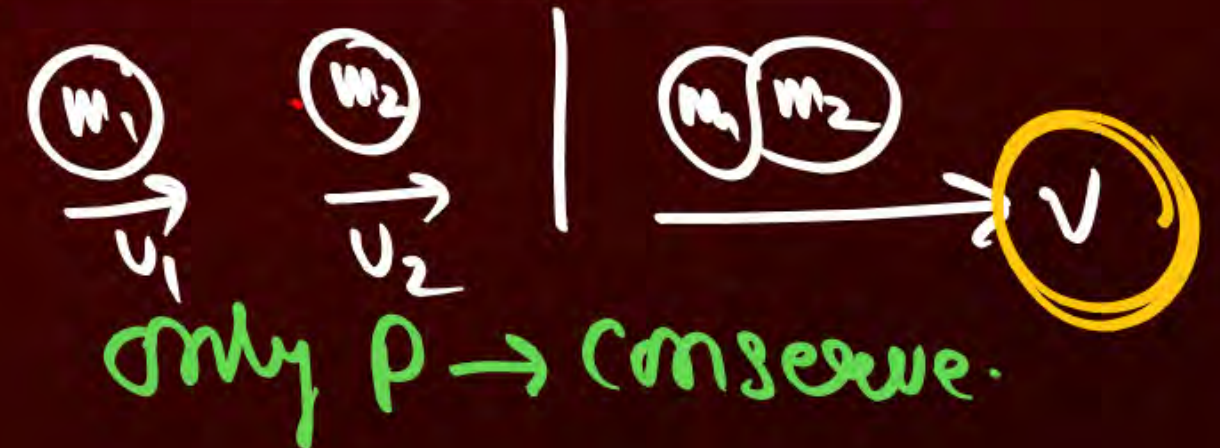
(Smart)

Collision

Perfectly elastic collision



Inelastic Collision

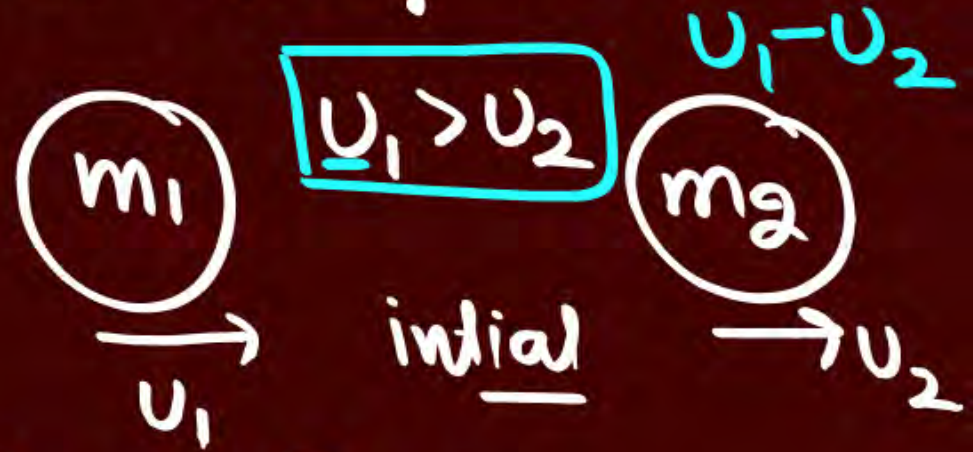


① Perfectly Elastic Collision

$\left. \begin{matrix} v_1 \\ v_2 \end{matrix} \right\}$ Baarhu ka kaam hai $\rightarrow$ lengthy

During elastic collision momentum and K.E both are conserved.

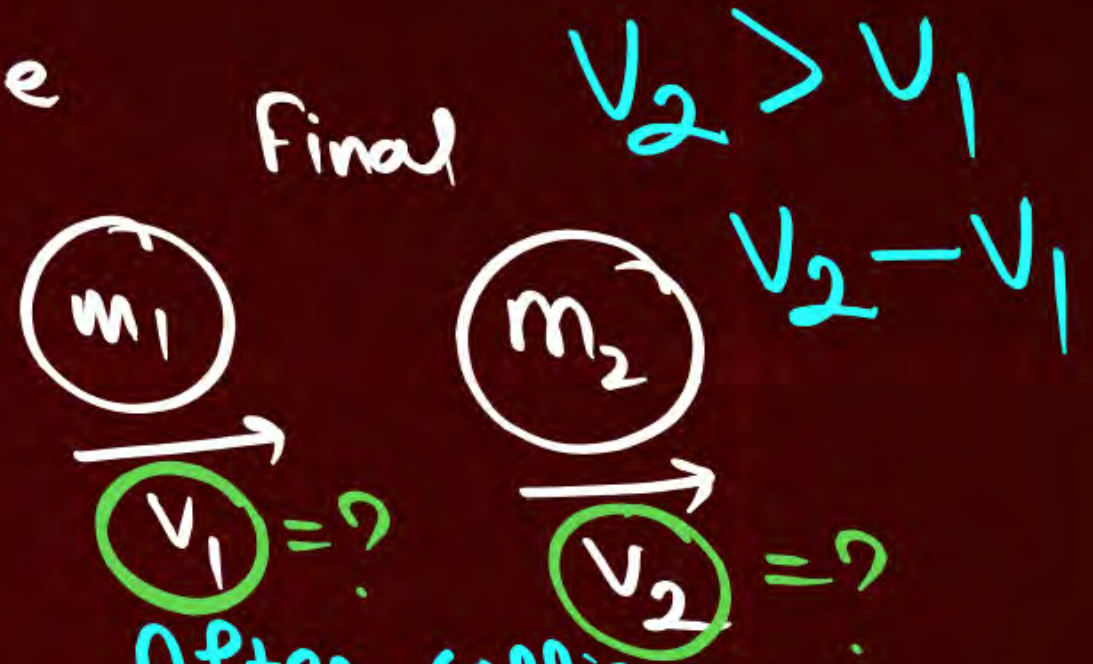
$\Rightarrow$ and Body donot change the shape



Before collision



During collision



After collision

② Momentum Conservation

$$P_i = P_f$$

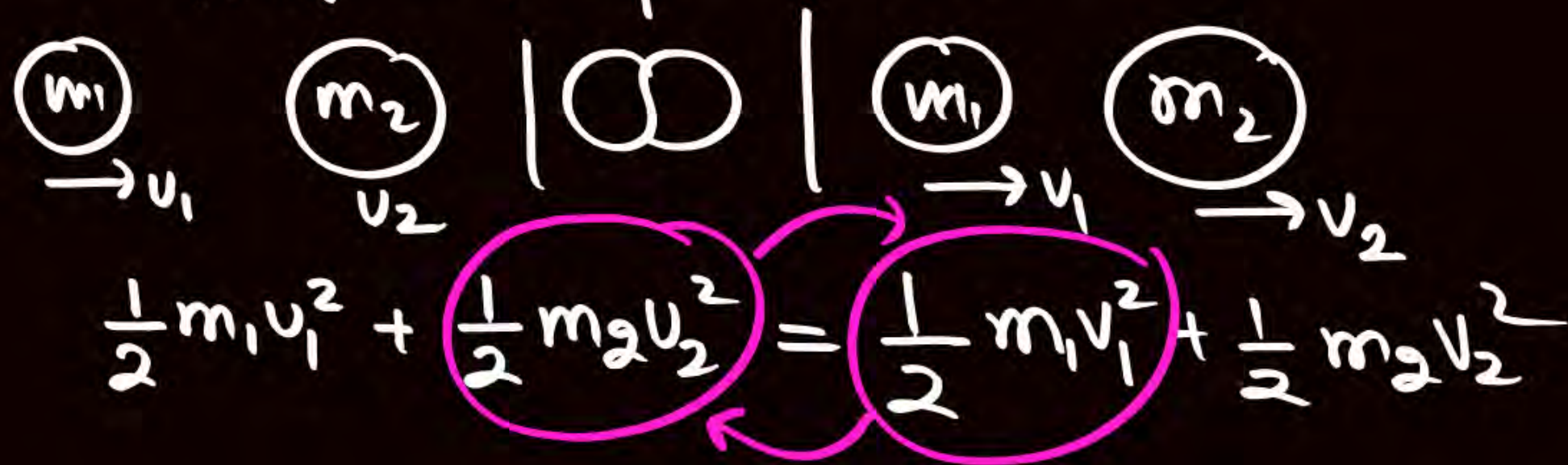
$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2 \quad \text{--- ①}$$

$$m_1 u_1 - m_1 v_1 = m_2 v_2 - m_2 u_2$$

$$m_1(u_1 - v_1) = m_2(v_2 - u_2) \quad \text{--- ②}$$

Kinetic Energy Conservation

$$K.E_i = K.E_f$$


$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

$$\underline{\frac{1}{2} m_1 u_1^2} - \underline{\frac{1}{2} m_1 v_1^2} = \underline{\frac{1}{2} m_2 v_2^2} - \underline{\frac{1}{2} m_2 u_2^2}$$

$$\cancel{\frac{1}{2}} m_1 (u_1^2 - v_1^2) = \cancel{\frac{1}{2}} m_2 (v_2^2 - u_2^2)$$
$$m_1 (u_1^2 - v_1^2) = m_2 (v_2^2 - u_2^2)$$

$$m_1 (\underline{u_1^2 - v_1^2}) = m_2 (\underline{v_2^2 - u_2^2}) \quad \# \quad a^2 - b^2 = (a+b)(a-b) \rightarrow \underline{\underline{6^{th}}}$$

$$m_1 (u_1 + v_1)(u_1 - v_1) = m_2 (v_2 + u_2)(v_2 - u_2) \quad \text{--- (III)}$$

Divide (III) and (II)

$$\cancel{m_1 (u_1 + v_1)(u_1 - v_1)} = \cancel{m_2 (v_2 + u_2)(v_2 - u_2)} \quad \text{--- (III)}$$

$$\cancel{m_1 (u_1 - v_1)} = \cancel{m_2 (v_2 - u_2)}$$

$$u_1 + v_1 = v_2 + u_2$$

$$\text{Velocity of approach} \quad u_1 - u_2 = v_2 - v_1 \quad \text{--- (IV)}$$

Velocity of approach

Velocity of separation

V_1 Final velocity of object



✓ From 1st equation

$$m_1 u_1 + m_2 u_2 = m_1 \underline{v_1} + m_2 \underline{v_2} \quad \text{Replace}$$

$$\# \quad u_1 - u_2 = \underline{v_2} - \underline{v_1}$$

$$\underline{u_1 - u_2 + v_1 = v_2}$$

JKBOSE

↳ Derivation

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 (u_1 - u_2 + v_1)$$

$$m_1 u_1 + m_2 u_2 = \underline{m_1 v_1} + m_2 u_1 - m_2 u_2 + \underline{m_2 v_1}$$

$$\underline{m_1 u_1 + m_2 u_2 + m_2 u_2 - m_2 u_1} = m_1 \underline{v_1} + m_2 \underline{v_1}$$

$$m_1 \underline{u_1} - m_2 \underline{u_1} + m_2 \underline{u_2} + m_2 \underline{u_2}$$

$$= v_1 (m_1 + m_2)$$

$$u_1 (m_1 - m_2) + 2m_2 u_2 = v_1 (m_1 + m_2)$$

$$\frac{u_1 (m_1 - m_2) + 2m_2 u_2}{(m_1 + m_2)} = v_1$$

$$\frac{u_1 (m_1 - m_2)}{(m_1 + m_2)} + \frac{2m_2 u_2}{(m_1 + m_2)} = v_1$$

$$V_1 = \frac{u_1 (m_1 - m_2)}{(m_1 + m_2)} + \frac{2m_2 u_2}{(m_1 + m_2)}$$

V₂ Final velocity of 2nd object



From 1st equation

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

replace

$$\# u_1 - u_2 = v_2 - v_1$$

$$v_1 + u_1 - u_2 = v_2$$

$$\boxed{v_1 = v_2 - u_1 + u_2}$$

$$\begin{array}{l} 2 + 4 = 6 \\ 4 + 2 = 6 \end{array}$$

$$m_1 u_1 + m_2 u_2 = m_1 (v_2 - u_1 + u_2) + m_2 v_2$$

$$\underline{m_1 u_1} + \underline{m_2 u_2} = \underline{m_1 v_2} - \underline{m_1 u_1} + \underline{m_1 u_2} + \underline{m_2 v_2}$$

$$m_1 u_1 + m_1 u_1 + m_2 u_2 - m_1 u_2 = m_1 v_2 + m_2 v_2$$

$$\underline{m_1 u_1} + \underline{m_1 u_1} + \underline{m_2 u_2} - \underline{m_1 u_2} = v_2 (m_1 + m_2)$$

$$2m_1 u_1 + u_2 (m_2 - m_1) = v_2 (m_1 + m_2)$$

$$\frac{2m_1 u_1 + u_2 (m_2 - m_1)}{(m_1 + m_2)} = v_2$$

$$\frac{2m_1 u_1}{(m_1 + m_2)} + \frac{u_2 (m_2 - m_1)}{(m_1 + m_2)} = v_2$$

$$\frac{u_2 (m_2 - m_1)}{(m_1 + m_2)} + \frac{2m_1 u_1}{(m_1 + m_2)} = v_2$$

Good

$$v_2 = \frac{v_2(m_2 - m_1)}{(m_1 + m_2)} + \frac{2m_1 v_1}{(m_1 + m_2)}$$

Inelastic Collision

During inelastic collision only momentum is conserved.

SUMMARY



- CF
- non-cf
- $VCM \rightarrow T$
 $\rightarrow CF$
- Collision
- PEC $v_1 \checkmark$
 $v_2 \checkmark$

HOMEWORK



Q $\left. \begin{matrix} v_1 \\ v_2 \end{matrix} \right\}$ Derivation Imp

Thank
You

